

Local installation of Firedrake, Python and ParaView

These instructions allow to install Firedrake on your computer and use the notebooks of the course **without needing internet connection**. At the end of this document you may also find some troubleshooting info on Paraview.

Firedrake works on **Linux** platforms: the instructions reported below regard a Linux installation.

- If you have **macOS**, you can try going directly through the same steps: you may find useful instructions regarding your system in the Firedrake pages linked below.
- If you have a **Windows** system, you need to install the *Windows Subsystem for Linux* (WSL). Instructions to install WSL can be found here: <https://learn.microsoft.com/en-us/windows/wsl/install>. Remember to **restart your computer** after the installation. To use WSL easily, we recommend that you *Install Windows Terminal* (find instructions and link at the same URL provided above).
Once you have installed WSL, you will be able to open a terminal with a full Linux distribution (except for some graphics features, that are however not needed for Firedrake): **in that terminal**, you can follow the instructions below.

Install Firedrake

The procedure, is mostly automatic up to point **5** (although point **3** requires interaction), but it may require up to **2-3 hours**.

To install Firedrake, follow the instructions at <https://www.firedrakeproject.org/download.html>. During the process, the following notes can be useful:

- 1** If you see any error in the early stages of the installation, try updating your system packages by `sudo apt update` and then `sudo apt upgrade`, then re-try installation.
- 2** The installation *may seem* to get stuck after

```
Creating firedrake venv in '/home/...../firedrake'.  
We use pybind11==2.9.2 until the issue with pybind11==2.10.0 is  
resolved.
```

→ just wait: it can take up to several minutes (typically less than 30 minutes).

- 3** If you see

```
The authenticity of host github.com ... can't be established`
```

answer **yes** to allow the download of the Firedrake code repository. You may probably see

```
Failed to clone ... using ssh, falling back to https.
```

many times: this is an issue related to the code repository hosted on github.com, but it will not affect your installation.

- 4 This message is self-explaining...

```
Installing petsc/  
Depending on your platform, PETSc may take an hour or more to build!
```

- 5 Activate the **firedrake** virtual environment created during installation by

```
source firedrake/bin/activate
```

In your terminal, the command line should then start with (**firedrake**).

- 6 Test your installation as indicated in the installation page <https://www.firedrakeproject.org/download.html>. If **pytest** is not found, install it inside the **firedrake** environment (i.e. run the following command in the same terminal where you executed point 5)

```
pip install pytest
```

and then re-try testing the installation:

```
pytest tests/regression/ -k "poisson_strong or stokes_mini or  
dg_advection"
```

Same if **vmtk** is not found.

Python notebooks on local machine

As an editor/interface for your python notebooks (*.ipynb files) plenty of options are available. We suggest Visual Studio Code, which is an IDE that you can download on your system just by visiting the page <https://code.visualstudio.com/>.

Then, we need to make your editor/IDE “talk” with the **firedrake** environment created above. Jupyter provides an easy notebook manager fitting this aim (more info on the Jupyter project at <https://jupyter.org/>). We install Jupyter **within the firedrake environment**, as follows:

- 1 Activate the **firedrake** environment:

```
source firedrake/bin/activate
```

- 2 Check that **pip** is up to date:

```
python -m pip install --upgrade pip
```

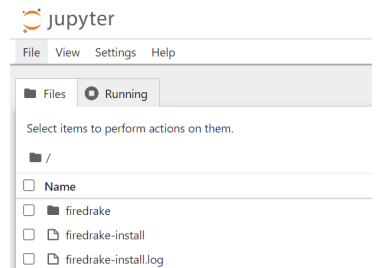
3 Install Jupyter

```
pip install jupyter
```

4 Launch a notebook server

```
jupyter notebook
```

- 5 The command at the previous point will print (possibly after a few seconds) two url's like
`http://localhost:8888/...`
`http://127.0.0.1:8888/...`
Copy one of them (they are interchangeable) and paste into your internet browser (even without internet connection!) to check that everything works. You should see something like this:



- 6 Configure VSCode to use this kind of environment: open VSCode, click in the top bar and select *Show and Run Commands* (or use the keyboard shortcut provided alongside this option), then start typing and select *Install Extensions*. By this, install the following extensions:
- (only for Windows) WSL
 - (for all systems) Jupyter, Jupyter Notebook, Python
 - if prompted, select the default versions/options.

Let's run the code!

The following instructions have to be run every time you start a programming session:

- 1 Open a terminal (WSL for Windows) and activate the **firedrake** environment created above:

```
source firedrake/bin/activate
```

- 2 Launch a notebook server within the environment:

```
jupyter notebook
```

- 3 Open your notebook editor and connect it to the python kernel on the running server. In VSCode:

1. On the top-right of your notebook you should see something like ... *kernel* ... or *base (Python ...)*: click on it.
2. *Select another kernel*→*Existing Jupyter Server...*
3. Copy the server url from the terminal and paste it here.
4. If prompted, select default options.

The same server can be used to open and run different notebooks: just re-execute this point for all the notebooks you open.

ParaView: installation and troubleshooting

At the following page you can find the executable and codes to install ParaView on your machine: <https://www.paraview.org/download>. Choose the latest version (NOT *nightly*) and, depending on your OS:

Windows Download and run the non-MPI installer, e.g.

`ParaView-6.0.0-Windows-Python3.12-msvc2017-AMD64.msi`

Linux Download and extract the tar archive. After extraction, you will find a `bin` folder containing the main executable `paraview`: this is what you need to launch.

macOS Check whether your processor is Silicon or Intel, then download either the pkg or dmg file (they are equivalent) and install them.

If you have troubles with your Windows installation, try the following (in order):

- 1 Try installing an older version: we suggest v5.10.1 or, if that still does not work, v.5.8.1.
- 2 Try fixing the version you have installed by following the indications suggested here: <https://discourse.paraview.org/t/how-to-run-paraview-including-the-binary-release-using-mesa-on-windows/4321>, where "ParaView_Path" is the folder where you have installed ParaView.
- 3 Try fixing the version you have installed by replacing the OpenGL library as indicated at <https://discourse.paraview.org/t/how-to-run-paraview-including-the-binary-release-using-mesa-on-windows/4321/8>, namely:
 - 1 from <https://fdossena.com/?p=mesa/index.frag> download the latest OpenGL (almost surely the 64-bit version, but check your processor!);
 - 2 extract it and copy the file `OpenGL3d.dll` into the `bin` subfolder of the folder where ParaView is installed.